



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT :

Network Design for Faculty of Computing Block N28c

TASK 5: IP ADDRESSING SCHEME

SECR1213 - NETWORK COMMUNICATIONS
SEMESTER I, SESSION 2024/2025
SECTION 02

Lecturer : Dr. ISMAIL FAUZI BIN ISNIN

NAME	Matric.No
TAN ZHI MING	A23CS0189
NG YU HIN	A23CS0148
LEE YIN SHEN	A23CS0236
MUHAMMAD NAIM BIN ABDULLAH	A23CS0134

Contents

5.1 Get the Network Address from your lecturer.....	2
5.2 Divide it in the best possible way for your network.....	3
5.2.1 Network distribution for different work area	3
5.2.2 Network distribution for workstation in different work areas.	6
Ground Floor :	6
First Floor:	13
Meeting Minute #1	18

5.1 Get the Network Address from your lecturer.

Group	Network IP address
1	10.16.0.0/12
2	10.32.0.0/12
3	10.48.0.0/12
4	10.64.0.0/12
5	10.80.0.0/12
6	10.96.0.0/12
7	10.112.0.0/12
8	10.128.0.0/12
9	10.144.0.0/12
10	10.176.0.0/12

Figure 5.1.1

Figure 5.1.1 shows the list of network addresses of each group. The network address of our group is 10.32.0.0/12. We assume that our subnet mask is /24 or 255.255.255.0 .

Network Address details:

IP address (decimal)	10.32.0.0
IP address (binary)	00001010 00100000 00000000 00000000
Subnet Mask (decimal)	255.255.255.0
Subnet Mask (binary)	11111111 11111111 11111111 00000000

We partitioned the subnet mask into a network portion (16 bits) and a host portion (8 bits). By performing the AND operation of the IP address with the subnet mask, we identified the network address as 10.32.0.0 and the broadcast address as 10.32.255.255. The valid address range within our building spans from 10.32.0.1 to 10.32.255.254.

Network Address	10.32.0.0
Broadcast Address	10.32.255.255
Range of useable address	10.32.0.1 - 10.32.255.254

5.2 Divide it in the best possible way for your network.

5.2.1 Network distribution for different work area

subnet	Network portion	Subnet bit	Host portion		IP address
0	00001010 0010	0000 00000000	00000000	Network address	10.32.0.0
			11111111	Broadcast address	10.32.0.255
1	00001010 0010	0000 00000001	00000000	Network address	10.32.1.0
			11111111	Broadcast address	10.32.1.255
2	00001010 0010	0000 00000010	00000000	Network address	10.32.2.0
			11111111	Broadcast address	10.32.2.255
3	00001010 0010	0000 00000011	00000000	Network address	10.32.3.0
			11111111	Broadcast address	10.32.3.255
4	00001010	0000	00000000	Network	10.32.4.0

	0010	00000100		address	
			11111111	Broadcast address	10.32.4.255
5	00001010 0010	0000 00000101	00000000	Network address	10.32.5.0
			11111111	Broadcast address	10.32.5.255
6	00001010 0010	0000 00000110	00000000	Network address	10.32.6.0
			11111111	Broadcast address	10.32.6.255
7	00001010 0010	0000 00000111	00000000	Network address	10.32.7.0
			11111111	Broadcast address	10.32.7.255
8	00001010 0010	0000 00001000	00000000	Network address	10.32.8.0
			11111111	Broadcast address	10.32.8.255
9	00001010 0010	0000 00001001	00000000	Network address	10.32.9.0
			11111111	Broadcast address	10.32.9.255
10	00001010 0010	0000 000010010	00000000	Network address	10.32.10.0
			11111111	Broadcast address	10.32.10.255

Subnet	Work area	Network address	Broadcast address	Range of usable address	Subnet mask address
0	Database room & Server room	10.32.0.0	10.32.0.255	10.32.0.1 - 10.32.0.254	255.255.255.0/24
Ground Floor					
1	General lab 1	10.32.1.0	10.32.1.255	10.32.1.1 - 10.32.1.254	255.255.255.0/24
2	General lab 2	10.32.2.0	10.32.2.255	10.32.2.1 - 10.32.2.254	255.255.255.0/24
3	Hybrid classroom	10.32.3.0	10.32.3.255	10.32.3.1 - 10.32.3.254	255.255.255.0/24
4	Voice conference room	10.32.4.0	10.32.4.255	10.32.4.1 - 10.32.4.254	255.255.255.0/24
5	Student lounge	10.32.5.0	10.32.5.255	10.32.5.1 - 10.32.5.254	255.255.255.0/24
First Floor					
6	NOC	10.32.6.0	10.32.6.255	10.32.6.1 - 10.32.6.254	255.255.255.0/24
7	ESL	10.32.7.0	10.32.7.255	10.32.7.1 - 10.32.7.254	255.255.255.0/24
8	CNL	10.32.8.0	10.32.8.255	10.32.8.1 - 10.32.8.254	255.255.255.0/24
9	MTR	10.32.9.0	10.32.9.255	10.32.9.1 - 10.32.9.254	255.255.255.0/24
10	MR	10.32.10.0	10.32.10.255	10.32.10.1 - 10.32.10.254	255.255.255.0/24

Ground Floor :



General Lab 1 :

Device	IP address
PC1	10.32.1.1
PC2	10.32.1.2
PC3	10.32.1.3
PC4	10.32.1.4
PC5	10.32.1.5
PC6	10.32.1.6
PC7	10.32.1.7
PC8	10.32.1.8
PC9	10.32.1.9
PC10	10.32.1.10
PC11	10.32.1.11
PC12	10.32.1.12
PC13	10.32.1.13
PC14	10.32.1.14
PC15	10.32.1.15
PC16	10.32.1.16
PC17	10.32.1.17
PC18	10.32.1.18
PC19	10.32.1.19
PC20	10.32.1.20
PC21	10.32.1.21
PC22	10.32.1.22
PC23	10.32.1.23

PC24	10.32.1.24
PC25	10.32.1.25
PC26	10.32.1.26
PC27	10.32.1.27
PC28	10.32.1.28
PC29	10.32.1.29
PC30	10.32.1.30
PC31	10.32.1.31
ACCESS POINT 1	10.32.1.32
Printer	10.32.1.33
Photostat Machine	10.32.1.34
BigScreenTV	10.32.1.35

General Lab 2 :

Device	IP address
PC1	10.32.2.1
PC2	10.32.2.2
PC3	10.32.2.3
PC4	10.32.2.4
PC5	10.32.2.5
PC6	10.32.2.6
PC7	10.32.2.7
PC8	10.32.2.8
PC9	10.32.2.9
PC10	10.32.2.10

PC11	10.32.2.11
PC12	10.32.2.12
PC13	10.32.2.13
PC14	10.32.2.14
PC15	10.32.2.15
PC16	10.32.2.16
PC17	10.32.2.17
PC18	10.32.2.18
PC19	10.32.2.19
PC20	10.32.2.20
PC21	10.32.2.21
PC22	10.32.2.22
PC23	10.32.2.23
PC24	10.32.2.24
PC25	10.32.2.25
PC26	10.32.2.26
PC27	10.32.2.27
PC28	10.32.2.28
PC29	10.32.2.29
PC30	10.32.2.30
PC31	10.32.2.31
ACCESS POINT 1	10.32.2.32
Printer	10.32.2.33
Photostat Machine	10.32.2.34
BigScreenTV	10.32.2.35

Hybrid Classroom :

Device	IP address
PC1	10.32.3.1
PC2	10.32.3.2
PC3	10.32.3.3
PC4	10.32.3.4
PC5	10.32.3.5
PC6	10.32.3.6
PC7	10.32.3.7
PC8	10.32.3.8
PC9	10.32.3.9
PC10	10.32.3.10
PC11	10.32.3.11
PC12	10.32.3.12
PC13	10.32.3.13
PC14	10.32.3.14
PC15	10.32.3.15
PC16	10.32.3.16
PC17	10.32.3.17
PC18	10.32.3.18
PC19	10.32.3.19
PC20	10.32.3.20
PC21	10.32.3.21
PC22	10.32.3.22

PC23	10.32.3.23
PC24	10.32.3.24
PC25	10.32.3.25
PC26	10.32.3.26
PC27	10.32.3.27
PC28	10.32.3.28
PC29	10.32.3.29
PC30	10.32.3.30
PC31	10.32.3.31
PC32	10.32.3.32
PC33	10.32.3.33
PC34	10.32.3.34
PC35	10.32.3.35
PC36	10.32.3.36
PC37	10.32.3.37
PC38	10.32.3.38
PC39	10.32.3.39
PC40	10.32.3.40
PC41	10.32.3.41
Access Point 1	10.32.3.42
Projector 1	10.32.3.43
Projector 2	10.32.3.44
BigScreenTV	10.32.3.45

Voice conference room :

Device	IP address
PC1	10.32.4.1
Projector 1	10.32.4.2
Photostat Machine	10.32.4.3
BigScreenTV	10.32.4.4
ACCESS POINT 1	10.32.4.5

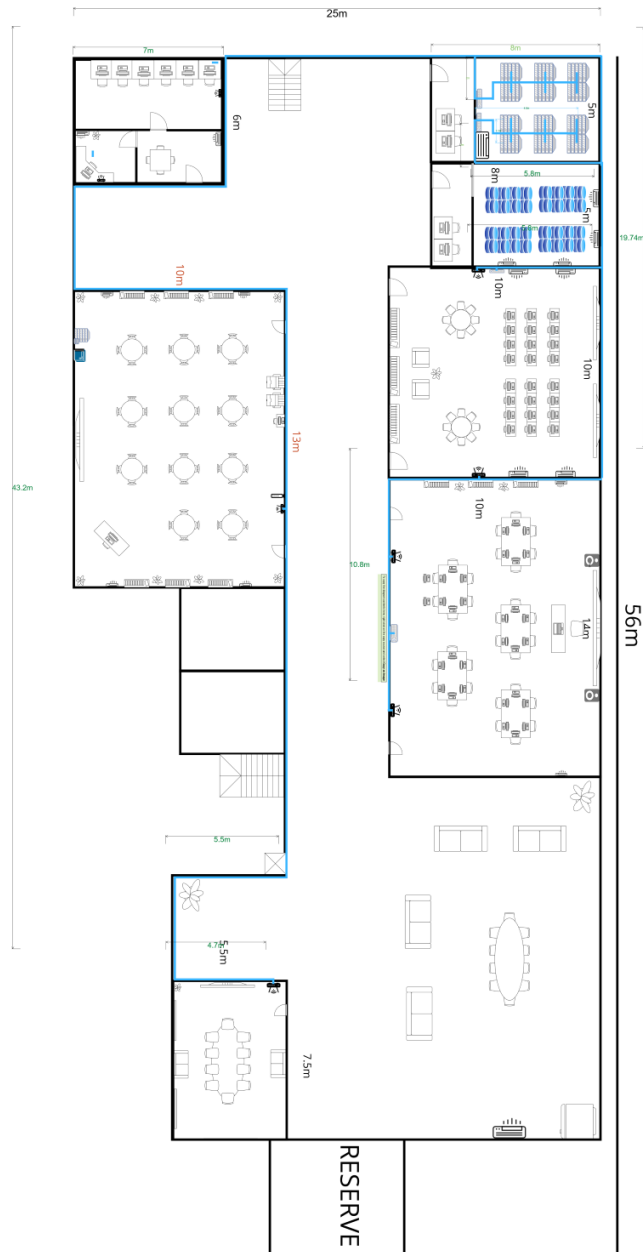
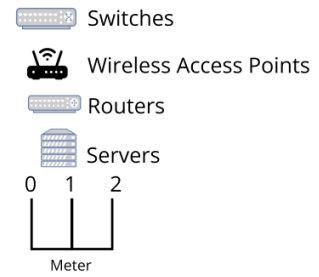
Student lounge:

Device	IP address
BigScreenTV	10.32.5.1

First Floor:



Indicators



NOC :

Device	IP address
PC1	10.32.6.1
PC2	10.32.6.2
PC3	10.32.6.3
PC4	10.32.6.4
PC5	10.32.6.5
PC6	10.32.6.6
PC7	10.32.6.7
PC8	10.32.6.8
PC9	10.32.6.9
PC10	10.32.6.10
PC11	10.32.6.11
PC12	10.32.6.12
PC13	10.32.6.13
PC14	10.32.6.14
PC15	10.32.6.15
PC16	10.32.6.16
PC17	10.32.6.17
PC18	10.32.6.18
PC19	10.32.6.19
PC20	10.32.6.20
PC21	10.32.6.21
PC22	10.32.6.22
PC23	10.32.6.23

PC24	10.32.6.24
------	------------

ESL :

Device	IP address
PC1	10.32.7.1
PC2	10.32.7.2
PC3	10.32.7.3
PC4	10.32.7.4
PC5	10.32.7.5
PC6	10.32.7.6
PC7	10.32.7.7
PC8	10.32.7.8
PC9	10.32.7.9
PC10	10.32.7.10
PC11	10.32.7.11
PC12	10.32.7.12
PC13	10.32.7.13
PC14	10.32.7.14
PC15	10.32.7.15
PC16	10.32.7.16
PC17	10.32.7.17
PC18	10.32.7.18
PC19	10.32.7.19
PC30	10.32.7.20
PC21	10.32.7.21
PC22	10.32.7.22

PC23	10.32.7.23
PC24	10.32.7.24
PC25	10.32.7.25
PC26	10.32.7.26
PC27	10.32.7.27
PC28	10.32.7.28
PC29	10.32.7.29
PC30	10.32.7.30
PC31	10.32.7.31
ACCESS POINT 1	10.32.7.32
BigScreenTV	10.32.7.33

CNL :

Device	IP address
PC 1	10.32.8.1
PC 2	10.32.8.2
ACCESS POINT 1	10.32.8.3

MTR :

Device	IP address
BigScreenTV	10.32.9.1

MR :

Device	IP address
ACCESS POINT 1	10.32.10.1

SR:

Device	IP address
Server 1	10.32.0.1
Server 2	10.32.0.2
Server 3	10.32.0.3
PC1	10.32.0.4
PC2	10.32.0.5

DR:

Device	IP address
PC1	10.32.0.6
PC2	10.32.0.7

Meeting Minute #1

Date/Time		20 DEC 2024 10.00 pm	
Location		Online (Google Meet)	
Agenda		1. Understand the details about task 5.	
		2. Discuss the way to divide the subnet.	
		3. Complete task 5 together.	
Meeting Host		Tan Zhi Ming	
Attendance			
Name		Time	Reason For Absence
Tan Zhi Ming		2200	-
Muhammad Naim Bin Abdullah		2200	-
Lee Yin Shen		2200	-
Eg Yu Hin		2200	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Discussion for task 5	- Zhi Ming shared the information about task 5 and its rubric.	Naim
		- All members read and investigate the task carefully.	
		- Solved the problem raised by group members together through discussion.	
2	Suggestion for subnetwork division	- All members suggest their prefer subnetwork division - All members discuss and decide the one that best suits	All members

		our design.	
3	IP Addressing Scheme Overview	- All members examine how IP addresses are currently assigned to each work area.	All members
4	Proceed to the parts of Task 5	- All members work on google docs together to complete task 5. - At the same time, if any of the group members question and identify any issues or problems then we find ways to encounter problems together.	All members
5	Meeting Ended	- At 0130, the meeting ended after all the discussion and work was done.	All members